

Coaching Guide to Major Ultramarathons

Never underestimate the value of course knowledge when it comes to racing an ultramarathon. The more you know about the terrain, weather, aid stations, gradients, durations of climbs, and landmarks, the better off you'll be on race day. Ultrarunning is an intellectual pursuit as much as it is a physical challenge, and having more knowledge enables you to create a better plan of attack.

Course reconnaissance is one way athletes gain knowledge about the demands of particular races, but that's not a realistic option for a lot of runners who travel significant distances to compete in their goal events. Some runners take the long view and race an event once as a recon mission with the idea they'll return a second time to pursue a performance goal. This is effective, but it's at least a two-year process! The most accessible way athletes learn about races is by talking with other athletes who have already done the event. As with training, however, this method suffers from the N-of-1 bias.

A benefit of coaching a large number of athletes over a long period is that I have been able to build a library of course-specific knowledge for major races. Based on input from athletes I work with, this coaching guide to major North American and international ultras is designed to supplement the information you'll find on the races' websites and in their race bibles. The goal here isn't to republish every detail about these well-known races but to provide the kind of insider advice and guidance that can only come from experience.

American River 50

The American River 50, held the first weekend of April, is one of the oldest and largest ultramarathons in the United States. The point-to-point course starts at Folsom Lake, east of Sacramento, California, and roughly follows the bike path around the lake and down the American River before doubling back on the other side and continuing into the Sierra Nevada foothills to the finish in Auburn, California.

The field of 800-plus runners enjoys a mostly flat course, especially for the first twenty-five miles. In fact, the race is frequently called “a road marathon followed by a trail run” thanks to its split personality. Experienced ultrarunners come to this race to notch PRs, while many newbies tackle it as their first ultra.

DID YOU KNOW? The American River 50 is known for one of the sweetest pieces of swag among ultra race finishers: a Patagonia jacket.

Course record: 5:32:18 (Jim Howard, 1981); 6:09:08 (Ann Trason, 1993)

Median time: 11:07

Cutoff time: 14 hours

Climbing:

Total elevation gain: 3,100 feet (half of which comes in the last five miles)

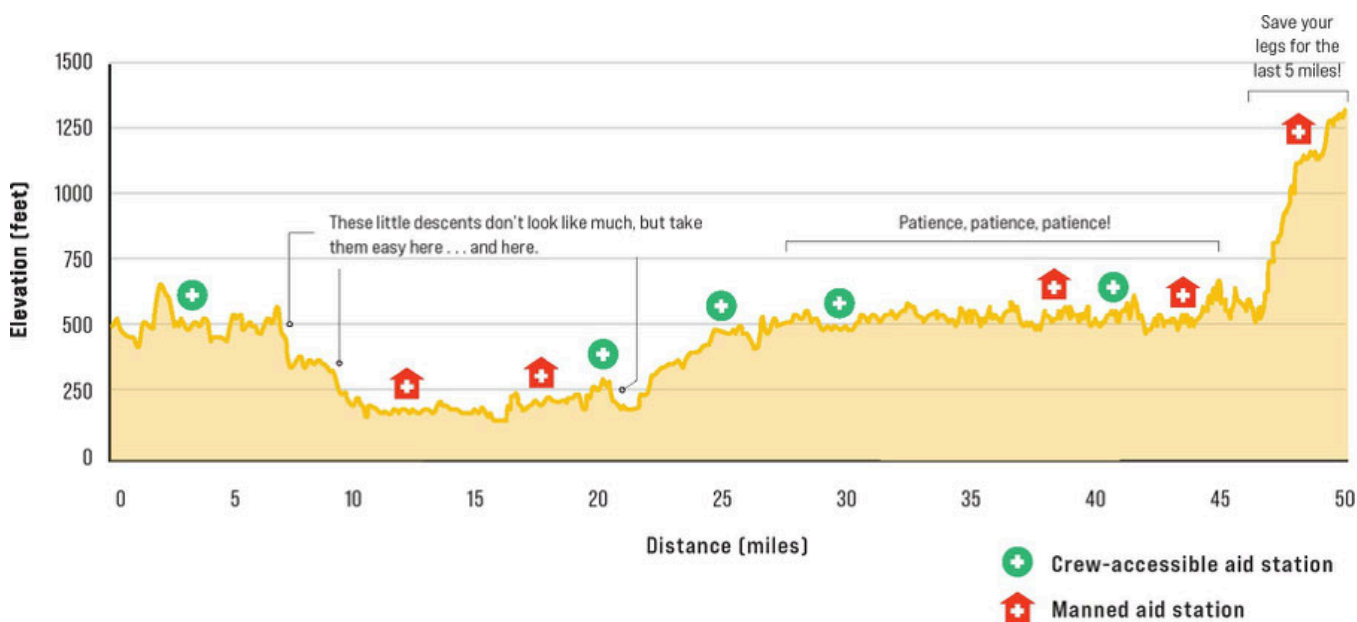
Total elevation loss: 2,100 feet

WEATHER. It's generally warm but not hot in the Sacramento Valley this time of year. The issue for many runners is that they aren't yet acclimated to the heat, even if that's only 75°F.

Rain is unlikely, but if it does rain, expect to spend the majority of the day in the wet because spring storms can take all day to move through the area.

UNIQUE WEATHER-RELATED CHALLENGE. The heat comes on in full force at the most difficult, final section of the race, which features the only long, sustained climb on the course over somewhat technical terrain (especially compared with the first half of the race).

EQUIPMENT. Despite the flat course and time spent on the bike trail, you'll want to wear trail running shoes to tackle the race's second half. You'll also need a headlamp for the first thirty minutes of the race, which you can then dump at the first aid station.



INDISPENSABLE GEAR. Carry two water bottles, as some aid stations are more than an hour apart.

CRUX OF THE RACE. The last five-mile section is a 1,500-foot climb that few are psychologically prepared for after cruising along for the first forty-five miles. The climb psyches many people out, so much so that some runners fail to finish the race even though they probably could.

CRITICAL MENTAL CHALLENGE. Runners who fail at the American River 50 tend to do so because they think it will be easier than it is. They find themselves loping along at marathon pace through the first couple of hours without even trying; they figure they're having a good day, not realizing that they're going too fast. Successful runners here have to force themselves to slow down during those first twenty-five miles, a challenging task when the field's size means there's always someone running faster around you.

TRAINING TIPS. You'll want to prepare to handle technical downhill sections. They're not long, but the race's relative speed can easily overmatch your agility—and state of fatigue—if you're not ready for them.

CRITICAL TRAINING PHASE. If you are using this race as a tune-up for your primary ultras later in the summer, feel free to do VO2 max work in the months leading up to the race. Although you will not tap into that intensity too much, it will set the stage for those more important upcoming races. If this is your goal event or your first ultra, you should be doing SteadyStateRun work in the four to eight weeks before the race.

CREW TIPS. Crew access is considered very easy. The main difficulty for your crew will be dealing with other crews when searching for parking space at the aid stations. As a result, crews may underestimate

travel and parking time and miss their runners. Plan out a strategy in which you only need to see your crew at certain, strategically chosen aid stations. This will be less stressful for you and them.

Pacers are allowed to join starting at mile 24.31, the Beal's Point aid station.

Badwater 135

This legendary 135-miler, which is run from the lowest and hottest point on the planet, Badwater, California, in Death Valley National Park (280 feet below sea level), to Whitney Portal at 8,300 feet, bills itself as “The World’s Toughest Foot Race” for four main reasons: it is run at the end of July when daytime highs regularly exceed 110°F; it is run entirely on bone-crushing tarmac; it crosses three mountain ranges; and it is thirty-five miles longer than the more popular 100-mile ultramarathons.

The field for Badwater is capped at 100 runners per year, and racers must qualify by completing three 100-mile races, with the most recent occurring within the preceding twelve months. Applicants who complete Badwater’s Salton Sea or Cape Fear races are given special consideration.

Course record: 21:33:01 (Yoshihiko Ishikawa, 2019); 24:13:24 (Patrycja Bereznowska, 2019)

Median time: 37:24

Cutoff time: 48 hours

Climbing:

Total elevation gain: 14,600 feet

Total elevation loss: 6,100 feet

Significant climbs:

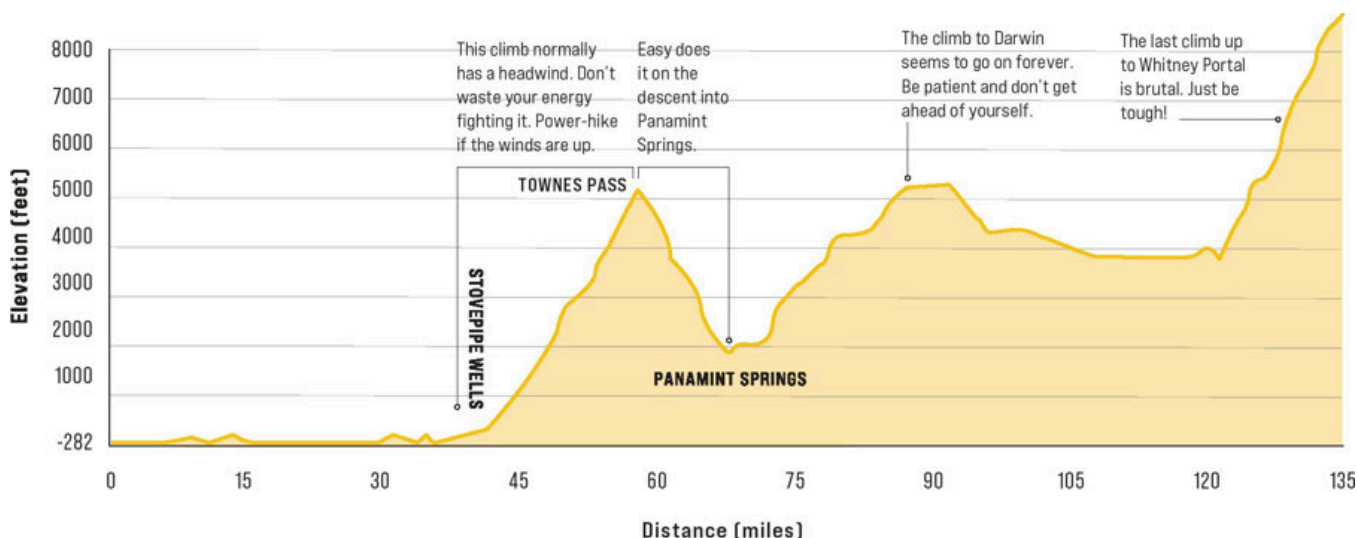
Stovepipe Wells to Townes Pass; Panamint Springs to Father Crowley Point; Lone Pine to Whitney Portal

DID YOU KNOW? The original Badwater race in 1977 finished atop Mount Whitney, the highest spot in the continental United States at 14,505 feet in elevation. That was an extra eight miles from (and 6,205 feet higher than) the current finish line at Whitney Portal.

WEATHER. Think you know heat? You don’t know heat until you try running in Death Valley, where the temperature can top 130°F, and it will stay above 100°F for significant portions of the race (minus a few

“cool” respites of 70°F temps on top of the mountain passes in the middle of the night). Despite the recent change to a night start, the course is still hot throughout the evening. There’s no rain, and you can pray for clouds, but don’t expect them.

UNIQUE WEATHER-RELATED CHALLENGE. It comes down to surviving the heat and the sun. Run on the relatively cooler white stripe on the shoulder of the road because the black asphalt will be hot enough to melt your shoes and blister the bottoms of your feet. There is usually a searing headwind, with gusts of twenty-five to thirty miles per hour, coming out of Stovepipe Wells at mile 41.5. It will suck the moisture right out of your nose and throat and make you choke. Consider applying saline solution inside your nose to keep it moist, and pop a steady supply of lozenges to help moisten your throat.



EQUIPMENT. With the extreme exposure and heat, proper clothing can make a big difference. You will want to wear clothes that will stay wet so you can enjoy some evaporative cooling effect. Therefore, avoid the fastest-drying and moisture-wicking fabrics. Ice sleeves are a must, as well as a hat that you can refill with ice every couple of miles.

INDISPENSABLE GEAR. Ice. Lots of ice. You’ll need more than you ever thought you would, and keeping it frozen inside an armory of high-quality coolers is a huge challenge. Not helping matters: there are very few places to find ice along the route, and everyone else in the race is looking for a lot of ice, too.

CRUX OF THE RACE. The entire race is probably unlike anything you’ve ever put yourself through, but the last half-marathon of the course from Lone Pine to Whitney Portal is a 4,573-foot climb straight up the side of the mountain. Coming at mile 122, it hits most racers at a point where they’ve already pushed beyond anything they’ve done before (run more than 100 miles).

CRITICAL MENTAL CHALLENGE. Because the ultrarunners who make up the field at Badwater are quite experienced, they are generally very good at heat acclimation and race-day planning. The time span and the sleep deprivation, not the heat, can be the hardest challenges. With the recent change to a night start, racers and crews can be out there for three consecutive nights, which is a first for many of them. Spending all that time in the race’s extreme conditions can leave even well-prepared racers battling through a complete breakdown of their bodies’ ability to thermoregulate. It’s not the temperature itself but rather the amount of time you’re exposed to it that eventually breaks you down.

THE PRO KNOWS. "Many ultrarunners can run all day and night on trails, but Badwater is a road race, not a trail race. Whereas a trail runner can do 75 percent of his training on dirt, training for Badwater means that more than 60 percent of your miles should be on pavement. It takes a long time to get the body used to that. I always tell people that running 200 miles anywhere else is nothing compared to finishing Badwater." —Dean Karnazes, ten-time Badwater 135 finisher

TRAINING TIPS. While for most hot-weather races a "minimum effective dose" strategy with heat acclimation is a good thing, Badwater is a different beast. Having a solid heat acclimation strategy is a good first step (described in depth in Chapter 7), but you will also want to do many of your training runs during the hottest parts of the day. This gives you the chance to test out some of your gear as well as familiarize your body with the rigors of running multiple hours in the heat.

CRITICAL TRAINING PHASE. Because the final few weeks of training should include critical heat acclimation, runners need to build their fitness and peak mileage six to eight weeks out from the race. Running your longest and hardest runs at the same time you are trying to acclimate to the heat is a recipe for disaster.

CREW TIPS. Your crew is vital to success in the Badwater 135—and also likely to suffer while offering assistance. In fact, more crew members than runners end up requiring medical attention for heat- and exhaustion-related emergencies. It's that brutal. Crew members need their own strict hydration and rest schedules to survive.

Successful teams need a minimum of two crew members to trade off duties, run resupply missions, and, most important, attend to their runner, who will need constant monitoring. Badwater is unique in that crews can help their runner at any time, anywhere on the course (except for certain sections of the race as specified by the race management). The only requirement is that they find a safe place to pull all four tires off the main road.

Pacers are allowed, but unlike other trail or road races, they can only run behind their runner, not in front or to the side.

Finally, read the rule book! No other ultramarathon in the United States, and perhaps the world, has more elaborate rules than Badwater. To protect the integrity of the race and ensure that future races happen, crews must be keenly aware of the rules.

Comrades Marathon

Held annually in June, the Comrades Marathon boasts the largest ultramarathon field in the world with approximately 25,000 runners participating annually. The renowned South African race is held on the roads between Durban and Pietermaritzburg in the KwaZulu-Natal province. The 55-mile point-to-point race alternates between “up” years and “down” years, yielding a net elevation gain or loss pending the direction. Due to the field size, the race infrastructure, and running on roads, the race will more closely resemble a large international road marathon than your run-of-the-mill trail ultra, complete with road running style aid stations spaced out every few kilometers. Should you choose to partake in the adventure, be prepared for enthusiastic crowds, starting corrals, and professional chip timing.

Course records:

Up: 5:24:39 (Leonid Shvetsov, 2008), 5:58:53 (Gerda Steyn, 2019)

Down: 5:18:19 (David Gatebe, 2016), 5:54:43 (Frith Van Merwe, 1989)

Median time: 10:42

Cutoff time: 12 hours

Climbing:

Total elevation gain: 6,300 feet for up years, 4,150 feet for down years

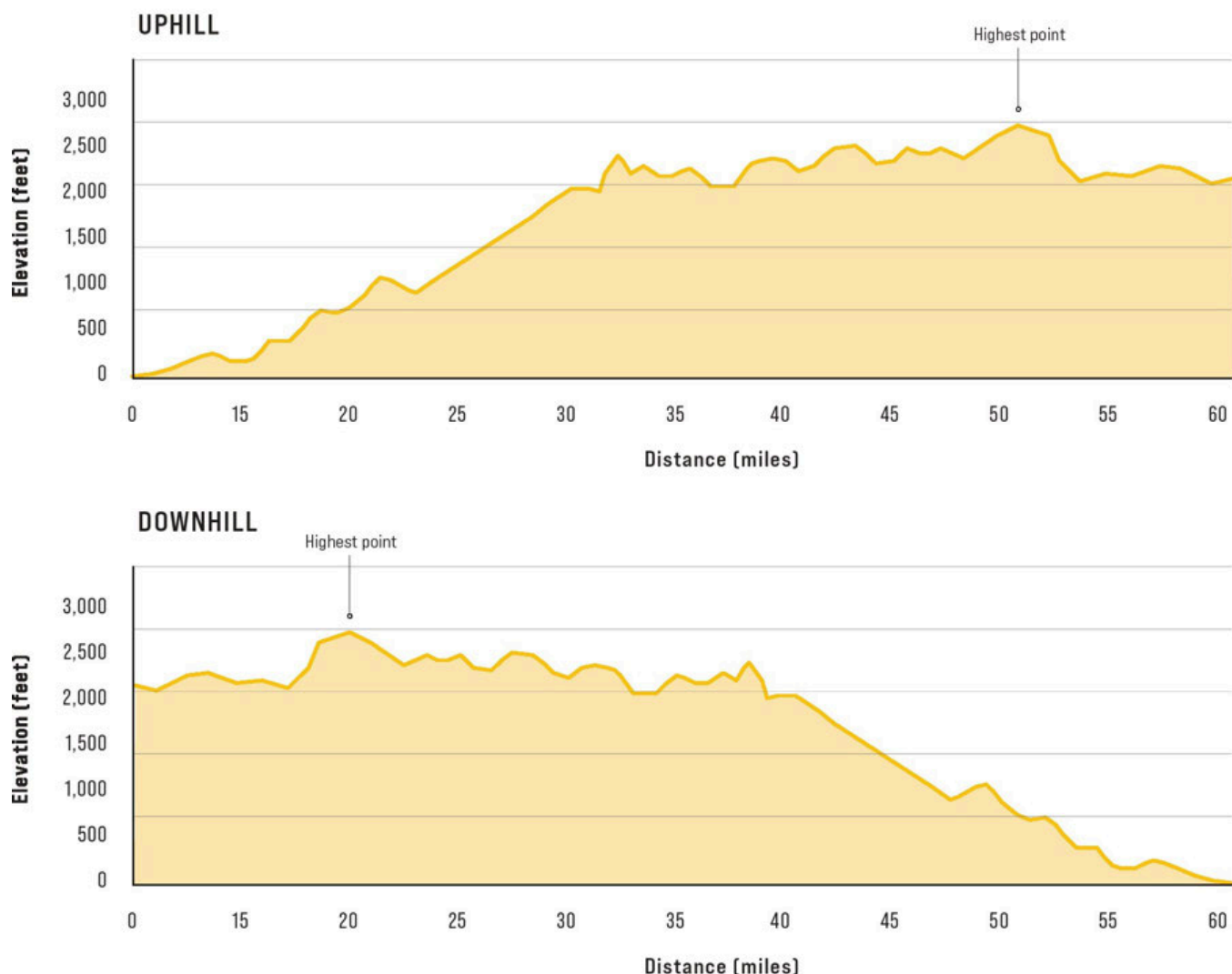
Total elevation loss: just the opposite of the gain!

DID YOU KNOW? Comrades was run for the first time in 1921, making it the oldest ultramarathon in the world. It has been run every year since, with the exceptions of during World War II and the COVID-19 pandemic. It attracts a stellar list of elite athletes competing for the richest ultrarunning prize purse (over \$250,000). The race awards cash prizes for the top ten overall finishers, top age group finishers, winners of the team competition, top South African runners, and top KwaZulu-Natal athletes.

HOW TO GET IN. Athletes have to qualify for the race by running strict time standards. Qualifying times can be from a marathon, various ultramarathon distances, and even the marathon leg of an Ironman® triathlon. Your qualifying time will determine your seeding position for the race.

WEATHER. The race is held in the middle of June and typically has favorable conditions. For runners from mountainous or cold-weather regions, it could be a touch warm. While no special heat acclimation is necessary, running in the afternoon in full sun for the last few weeks can help.

Aid stations are located every 2k of the race (43 in total)



CRITICAL MENTAL CHALLENGE. The atmosphere of the race is more like a road marathon than a trail ultra. It is easy to get sucked into the crowd in the early miles and run too hard. Particularly in a downhill year, where your early pacing strategy should revolve around saving your legs, runners should pay attention to a pacing plan based off of their predicted finish time. This is an instance where relying on your watch and splits might be of more benefit than perceived exertion.

TRAINING TIPS. On uphill years, training for the Comrades can be similar to training for any flatter 50-mile or 100K trail ultra. April and May should consist of TempoRuns and maybe SteadyStateRuns for the back-of-the-packers. If you are normally a trail runner, running specifically on roads three times per week will need to be added in during the last several weeks. For runners training for a downhill year, realize that your race pace during portions of the event will be faster than your normal EnduranceRun pace or even SteadyStateRun pace. Because of this, you can do TempoRuns right up to the taper of the event. If you regularly train on trails, make sure you are picking more benign trails so you don't lose any footspeed. If you choose to do specific downhill work, sprinkle it in lightly; three or four sessions total during the last six weeks will be plenty.

For some mid- to back-of-the-packers, there will be some climbs that are more effective to walk vs run. Don't let the stigma of "walking on the road" get to you. If it's more effective to power-hike, then do it and save your legs for later in the race.

Hardrock 100

One of the toughest 100-mile races in the world, the Hardrock, held in mid-July, starts and finishes in the high alpine town of Silverton, Colorado, and sends competitors on a loop around the most remote and highest mountain passes in the state, including an ascent up and over 14,048-foot Handies Peak. The route's elevation averages 11,186 feet, topping 12,000 feet thirteen times.

Getting into the race is almost as difficult as finishing it. Runners must have completed a qualifying ultra run, and even then there are only forty-seven slots available for first-time participants. In addition, there is a service requirement: eight hours of volunteer work at an ultra race.

Course record: 22:41 (Kilian Jornet, 2014); 27:18 (Diana Finkel, 2009)

Median time: 40:31

Cutoff time: 48 hours

Climbing:

Total elevation gain/ loss: 33,050 feet

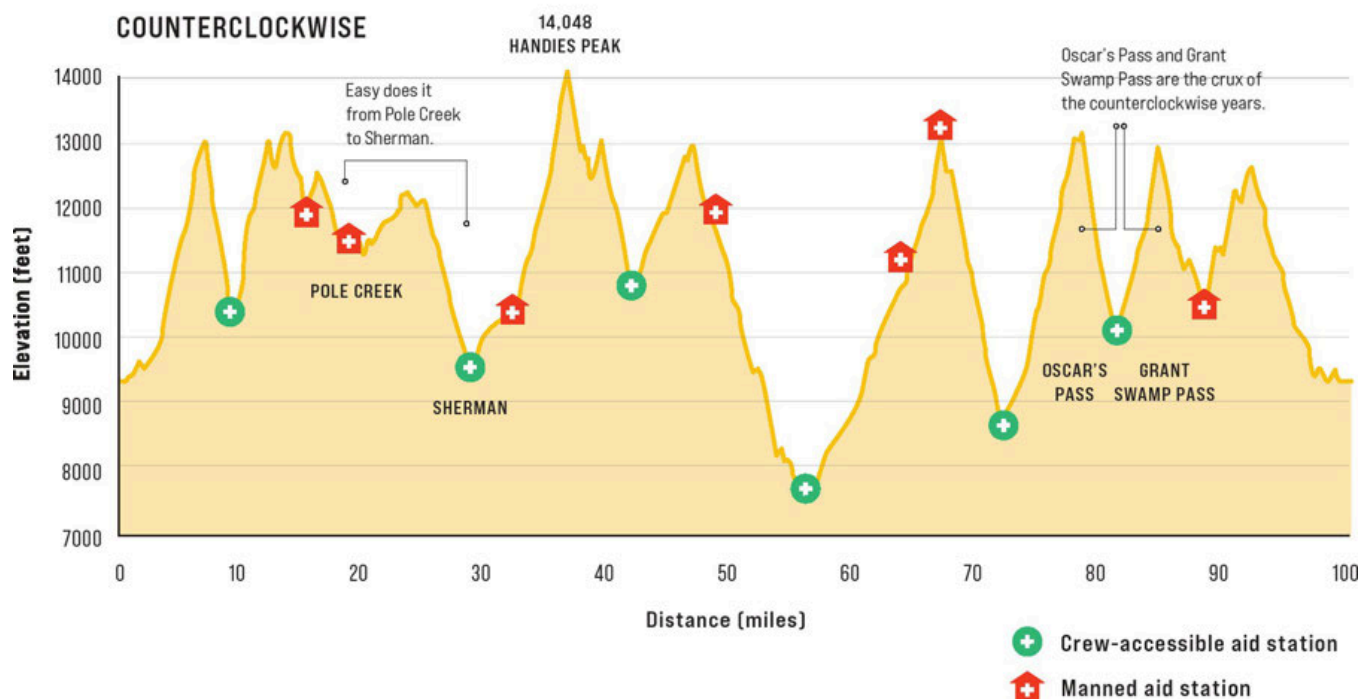
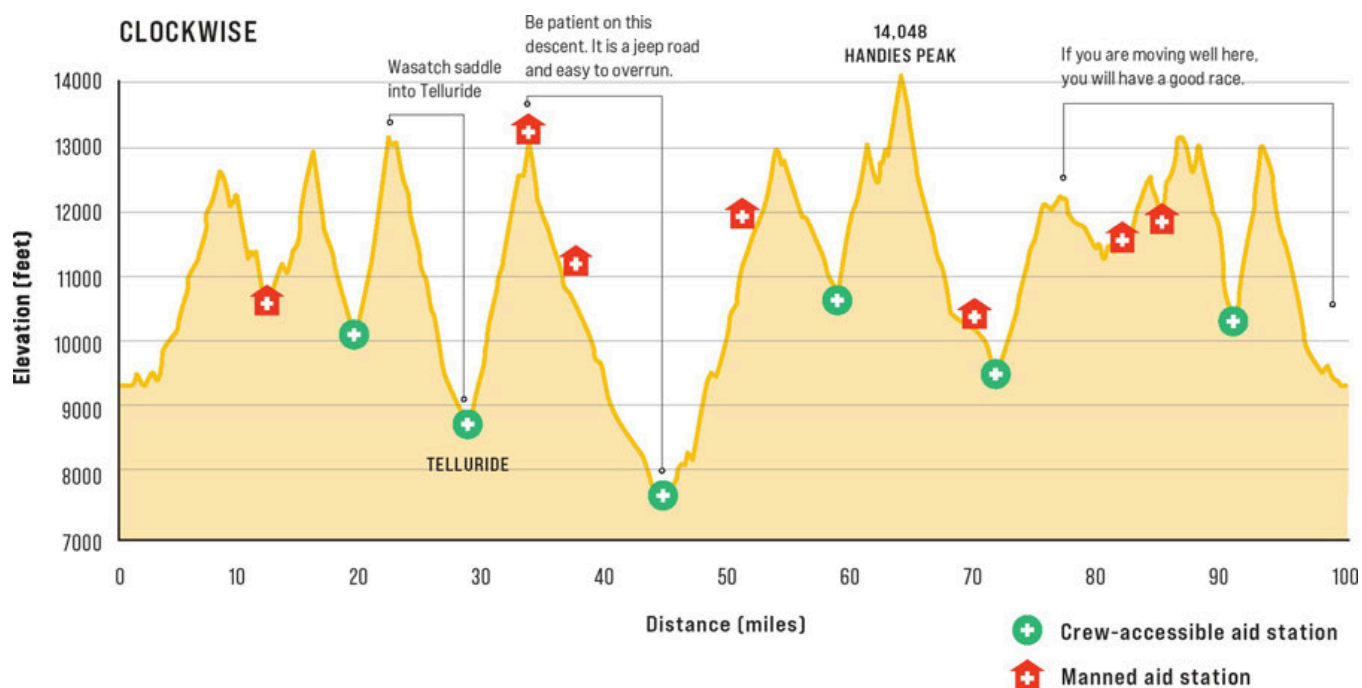
Significant climbs: They are all hard! Hardrock is one of the most notoriously difficult 100-mile events. You are climbing or descending nearly the entire course.

DID YOU KNOW? On even years the race goes clockwise. On odd years it runs counterclockwise. Racers have to kiss the Hardrock, a large stone painted with a picture of a ram's head, to officially finish the race. This race also has one of the most detailed course descriptions available; read it.

WEATHER. Prepare for subfreezing nights in the high alpine sections and highs in the mid-70s in the town of Ouray. Because the race takes place in the middle of the Rocky Mountain summer monsoon season, expect late afternoon thunderstorms, some of them violent, with extreme drops in temperature and even snowfall. Fortunately, the storms move in and out quickly most of the time.

UNIQUE WEATHER-RELATED CHALLENGE. Be on the lookout for clouds developing into storms that can unleash a bone-chilling cloudburst of rain or hail. The route takes participants over multiple exposed

ridgelines where lightning strikes are a very real concern. If you see or hear lightning nearby, work your way down to a lower elevation as quickly and safely as possible.



THE PRO KNOWS.

“On years when the race runs counterclockwise, you have to watch your pace through the Pole Creek section. You can easily let yourself fly down the course to the Sherman aid station and blow your whole race. Same goes for

the descent off the Wasatch saddle into Telluride in the clockwise years. People get so excited to see the town below that they blow out their quads by going too fast.”

—*Missy Gosney, fourth place, 2015*

EQUIPMENT. Carry a packable waterproof, breathable rain jacket at all times. Wrap your mind around the fact that you might be out on the course for two nights, and pack accordingly. Pack a complete change of clothes in every drop bag (from underwear to outerwear). Trekking poles can help steady your pace on the downhills.

INDISPENSABLE GEAR. An emergency space blanket will take up hardly any space but may prove a godsend if you need to wait out a storm before heading over a pass.

CRUX OF THE RACE. Handies Peak, at more than 14,000 feet, is higher than many people have ever climbed before, much less in the middle of the night, which will be the case for many runners during clockwise years. The altitude can slow you down more than you expect, but pushing too hard at this altitude can take a lot out of you, so gauge your effort by exertion rather than speed.

CRITICAL MENTAL CHALLENGE. It’s a long race. A good time is under thirty hours, which is the cutoff for other less challenging 100-milers. Even for experienced ultrarunners, the Hardrock will have you on your feet longer than you are used to and running at higher elevations than you may be comfortable with.

The course isn’t particularly well marked compared with other races, and that is by design. Racers should know the course before showing up at the start line. During the race, participants (and their pacers) must pay attention to their whereabouts at all times.

TRAINING TIPS. Training for the Hardrock is about banking as much vertical in the legs as possible. Shoot for long runs with 4,000 to 5,000 feet of climbing/descending on successive days. And take any chance you have to run at altitude, not necessarily to acclimate but to understand how your body responds to it.

CRITICAL TRAINING PHASE. May and June, with lots of hiking and descending. Most athletes cannot match the 680 feet of elevation change per mile in training, but it is critical to do whatever you can. Even for the winners, nearly all the uphill will be hikes, so spend the majority of May and June hiking as much as possible.

CREW TIPS. Crews are allowed only at the designated aid stations, some of which are remote with no cellular phone service. Crew members need to prepare for a lot of driving on twisting mountain roads and Forest Service roads.

Pacers need to make sure to pack for a long time on the trail, as much as ten hours, and carry enough water and food accordingly. Because there are few, if any, bailouts between aid stations, pacers must understand that they have to be prepared to go the distance.

Javelina Jundred

The setup for the Javelina Jundred makes it unusual among ultras. The popular race with more than 600 participants takes place entirely inside McDowell Mountain Regional Park, which is situated just outside the sprawl of Scottsdale, Arizona. The race, either a 100K or a 100-miler, is relatively flat and consists of twenty-mile loops in the park. What's unique about the race is that with each lap, runners reverse direction to run the next loop, giving everyone multiple chances to see each other (and for those aiming to win the race, a chance to see where their competition is).

Relatively little elevation change, easy logistics, and timing at the very end of the running season make the Javelina Jundred attractive to ultrarunners looking to qualify for the Western States 100-Mile Endurance Run.

Course record:

100-mile: 13:01:14 (Patrick Regan, 2017); 14:52:06 (Devon Yanko, 2015)

100K: 7:58:43 (Christian Gering, 2019); 8:48:25 (Courtney Dauwalter, 2016)

Median time:

100-mile: 26:01:26

100K: 16:28:57

Cutoff time:

100-mile: 30 hours

100K: 29 hours

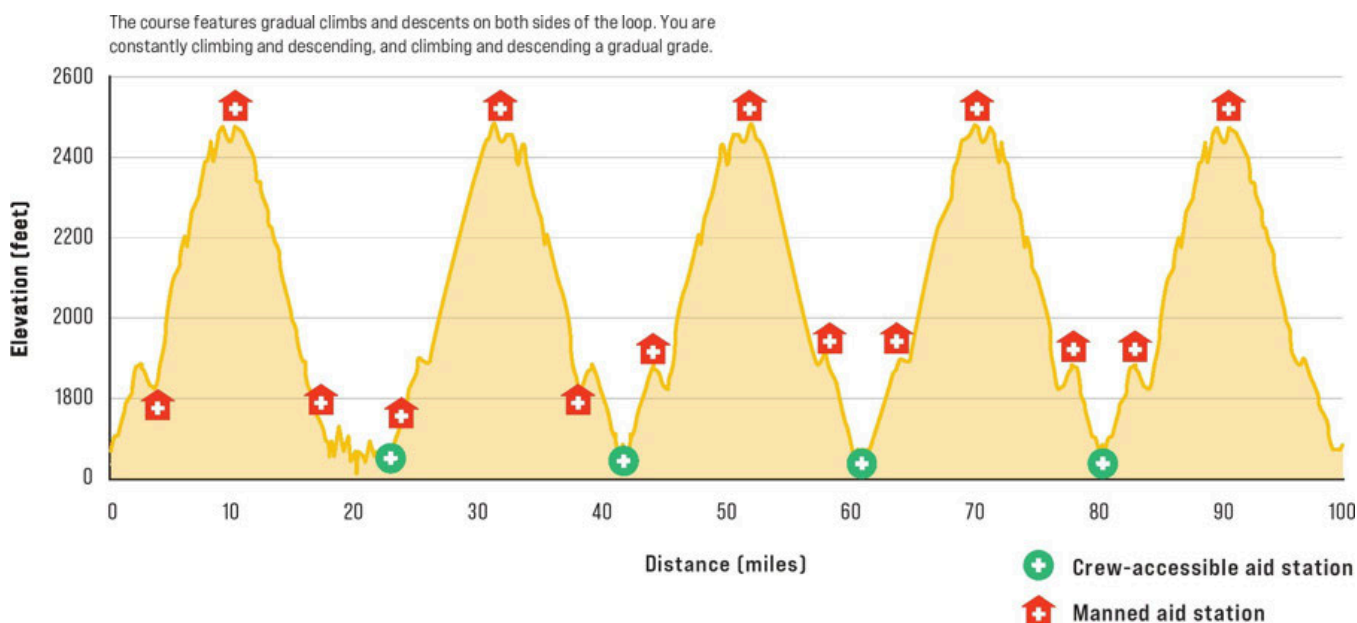
Climbing:

Total elevation gain/loss: 100-mile: 7,900 feet. 100K: 5,000 feet

DID YOU KNOW? Because the race occurs on or near Halloween, costumes are encouraged for runners and their crews, with awards given for the best male and female costumes. And because crews are allowed only at the start/finish line, they set up camp for the duration of the race and turn it into an all-day/all-night party. I have run as a Chippendale dancer.

WEATHER. As far as ultras go, the weather is relatively mild, with overnight temperatures in the high 50s to low 60s. Daytime highs can reach the mid-90s. Rain is rare, but it can happen.

UNIQUE WEATHER-RELATED CHALLENGE. The timing of the race at the end of October means that runners from farther north have already spent months running in much cooler fall weather. When they arrive in Arizona, they're not acclimated to running in 90-degree sun, with no shade anywhere on the course. As a result, expect to run slower than you may have planned due to the heat. Sunset doesn't always bring relief either. Many runners (and crews) aren't prepared for the sudden 40-degree drop in temperature once the sun goes down.



EQUIPMENT. With minimal weather-related issues and regular aid station intervals (you're never more than six-and-a-half miles from the next aid station), there's no need to carry any special equipment beyond a water bottle or two. The race start/finish line is part aid station, part Woodstock with runner crews and their tents lining the course. So, have your crew bring a pop-up tent and some chairs to enjoy the event. Everyone needs to shuttle in from an off-site parking area.

INDISPENSABLE GEAR. Many runners wear gaiters to keep the grit and small pebbles from the sandy washes and trail out of their shoes.

CRUX OF THE RACE. The imperceptible descent from the backside of the course to the start/finish line lulls many runners into thinking they're going to have an A+ day, and they start running faster than they should. When they start the slog back up that incline, they pay the price. Although there is not a lot of elevation gain/loss, you are very rarely running on perfectly flat terrain.

CRITICAL MENTAL CHALLENGE. Despite its easy logistics and relatively flat course, the Javelina Jundred is a deceptively hard race, with a finishing rate of only roughly 50 percent. The reason, beyond the heat, is that with every twenty-mile lap, racers return to where all the crews' camps are set up. During the night, the start/finish turns into a party atmosphere, with music, beer trucks, disco balls—it's a hard environment to leave. Even the backside aid station is set up as a party. Help yourself by not sitting down and getting sucked into the scene.

THE PRO KNOWS.

"Many people start out this race too fast because of the cool morning temperatures and easy geography. After the first lap, at least a third of the field is usually running a sub-twenty-four-hour pace, which is unsustainable unless they're an elite pro. The key to finishing is to force yourself to go slow during the first couple of laps."

—*Jamil Coury, race director*

TRAINING TIPS. If you live in a climate that features cool, crisp autumns, train during the heat of the day, not the mornings or evenings. Get comfortable running in sand and gravel—dry riverbeds work well. You don't need to do all of your runs in sand and gravel, but do enough to get used to them. Thanks to the terrain, it's possible to run the entire Javelina Jundred course, which makes pacing discipline a must.

CRITICAL TRAINING PHASE. Both the 100K and the 100-mile Javelinas are steady grinders, so EnduranceRun and SteadyStateRun intensity will be your bread and butter in September and October.

CREW TIPS. There may be no easier race to crew than the Javelina Jundred. The hardest part is shuttling in all your gear and camping equipment the afternoon or evening before the event starts. But after setting up camp, the crew doesn't need to move. Accordingly, make sure they—and you—have enough food, fluids, and creature comforts to last for the entire race, since they can't easily hop in a car to go get supplies or a meal.

As soon as the sun goes down on day one or after the third loop, whichever comes first, pacers are allowed to join their runners to the finish of the race.

JFK 50

The oldest ultra run in America, the JFK 50 started in 1962. The point-to-point race, held every November between Boonsboro and Williamsport, Maryland, is now the largest ultra in the country, with a cap of 1,000 runners. The course starts in western Maryland on the Appalachian Trail before connecting with the sublime C&O Canal Towpath along the Potomac River on the way to the finish. It's considered a fast and relatively easy course without much climbing (and nearly all the climbing is completed in the first five miles). But it's a different kind of race, as champions of western 100-mile races soon discover when they fail to crack the top ten.

Course record: 5:21:28 (Jim Walmsley, 2016); 6:11:59 (Ellie Greenwood, 2012)

Median time: 10:20

Cutoff time: 13 hours

Climbing: Total elevation gain/loss: 2,077 feet

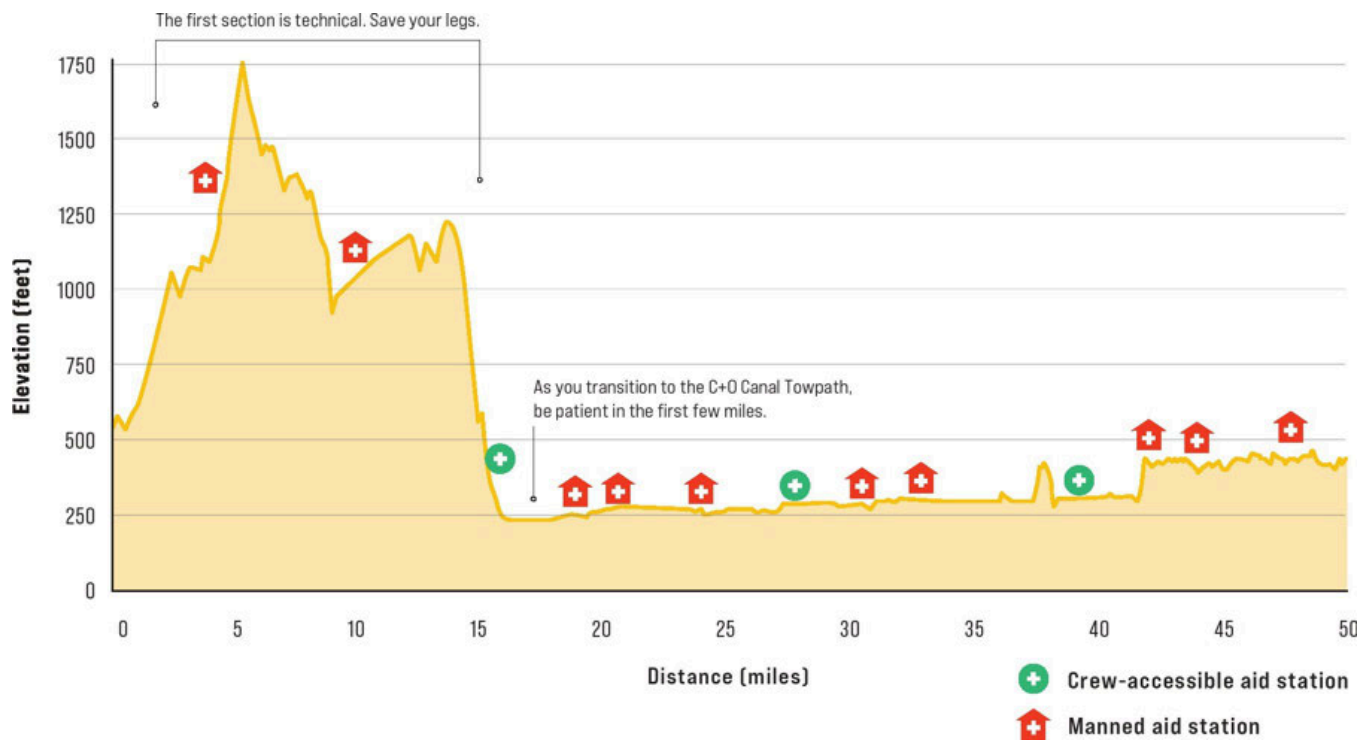
DID YOU KNOW? The 50-mile distance came from a challenge to the US Marines by then president John F. Kennedy to hike fifty miles in under twenty hours, as Teddy Roosevelt's marines had done. That led a group of eleven civilians to try it on what is now the JFK 50 course. By 1970, seventy-three participants had finished. For comparison, the first New York City Marathon held in 1970 recorded only fifty-five finishers.

WEATHER. Expect crisp fall weather for the Appalachian Mountains: highs in the mid-50s and lows at start time in the high 30s. It can rain, though. It can even snow.

UNIQUE WEATHER-RELATED CHALLENGE. It's rarely uncomfortably hot for this race; it's usually a glorious running day with perfect temperatures. Or it could rain, sleet, or snow. Prepare for the latter by stocking your drop bags with plenty of dry clothes to change into.

INDISPENSABLE GEAR. A space blanket is a small insurance policy worth carrying in case your day goes south and the cold starts getting to you.

CRUX OF THE RACE. The majority of the climbing takes place within five miles of the start, with a lot of it on technical, rocky trail. It will be a conga line up and down the mountains. If you're not careful, this can become a giant energy suck as you're constantly trying to sprint around people on the trail. If you find yourself stuck in the pack, save your energy, settle in, and carefully pick spots where you can pass more people more easily.



CRITICAL MENTAL CHALLENGE. Once runners hit the flat, soft gravel of the C&O Canal Towpath, many unwisely figure the hard part's over and pour on too much speed in those first few towpath miles. Holding yourself in check and sticking with your race plan is critical here; you still have thirty-five miles to go.

TRAINING TIP. Except for parts of the roughly ten-mile stretch along the Appalachian Trail, the JFK 50 is a completely runnable event, and you can easily settle into a rhythm for the duration of the course. You have to train for the trails and for the flats. Versatility is key for success in this event.

CRITICAL TRAINING PHASE. This is a late-season race for many ultrarunners, so starting healthy is a priority. Training during July, August, and September should include TempoRuns and SteadyStateRuns, which will pay off in this race.

CREW TIPS. The horseshoe-shaped route of the race makes it relatively easy for crew members to support their runners. That said, with 1,000 participants, your crew will likely find themselves more stressed out over the battle for parking at aid stations than over helping you, especially for the first few aid stations when the field will still be relatively bunched up. Better to plan on a few strategic meeting points and a smart drop bag strategy for the other points to get you through the race.

Pacers are not encouraged, but they are allowed.

Lake Sonoma 50

This early-season 50-miler is held every April in the beautiful and lush (at that time of year) Sonoma Valley in Northern California. The out-and-back rolling trail course is considered fast and features many sections through forest shade, across twelve creeks, and around its namesake lake. Trail runners love it; 86 percent of the course is on relatively smooth, nontechnical singletrack trails. The race has a high finishing rate (approximately 90 percent) and is often used as a tune-up race for the Western States 100, which is held in June. Entry is via a blind lottery (i.e., there's no preferential treatment for past entrants or pros).

Course record: 5:51:16 (Jim Walmsley, 2018); 7:08:23 (Stephanie Howe, 2015)

Median time: 11:03

Cutoff time: 14 hours

Climbing: Total elevation gain/loss: 10,500 feet

DID YOU KNOW? The Island View aid station on the course is so remote that supplies and volunteers have to be delivered by boat.

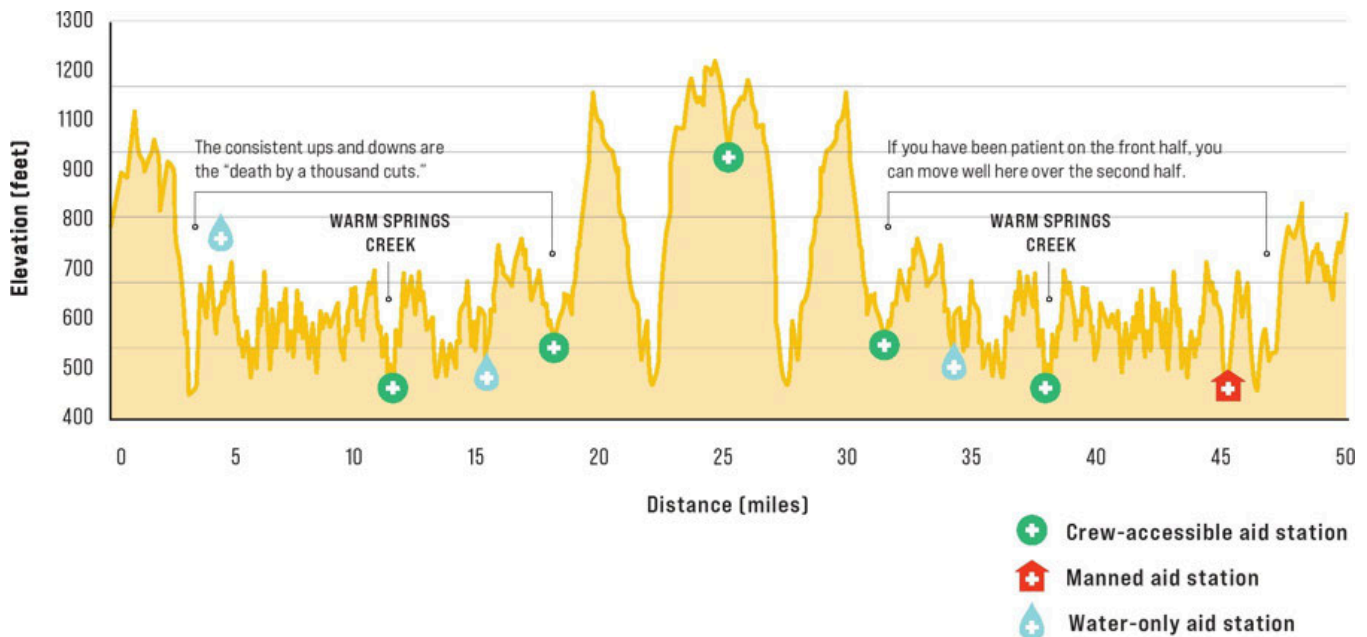
WEATHER. April in the Sonoma Valley has near-perfect ultrarunning weather, with lows in the mid-40s and highs in the mid-70s. Even during the heat of the day, much of the trail is under the shade of tree cover. There's one caveat (it's a big one): if it's raining, there's a good chance that it will rain hard throughout the day and that the temperature will stay uncomfortably cool.

UNIQUE WEATHER-RELATED CHALLENGE. None, really. It's California. It's spring. Odds are it's going to be a nice day.

EQUIPMENT. There are no special equipment needs and no need to carry much beyond water and food. Also, water at the more remote aid stations will be limited to drinking, not dousing yourself to cool off; you'll want to do that at the creek crossings or at the lake.

INDISPENSABLE GEAR. Some aid stations are nearly seven and a half miles apart, a long distance during the hottest part of the race. As such, carry more water and food than you think you might need.

CRUX OF THE RACE. From mile 33 to the finish, the race can feel long and lonely, with roughly five to seven and a half miles between the last three aid stations. Not helping matters, you won't see many people due to the terrain and how spread out everyone is. For midpackers and those at the back, the race can suddenly feel like running through the middle of nowhere by yourself.



THE PRO KNOWS.

"To do well in this race, you've got to be able to run fast, yet deal with a lot of climbing. That combination is unique in the ultra world, which is why you want to train for running this race— not just speed-hiking— uphill and down."

—*Dakota Jones, first place, 2012*

CRITICAL MENTAL CHALLENGE. For a 50-mile race with no mountains to climb, the Lake Sonoma 50 packs a serious dose of vertical into its punchy course. The relentless ups and downs with few flat sections or gradual inclines to rest can blow racers' quads apart well before the finish. The key, as with any ultra, is pacing, especially on the short downhill sections. "Death by a thousand cuts" is how former course record holder Alex Varner describes the multitude of ups and downs.

TRAINING TIPS. Work on your foot turnover to maintain your speed on the short flats and to mitigate some of the punishment (but not speed) on the downhill stretches. The race requires a lot of pace changes; work on that in training by including ups, downs, and flats in any specific interval work.

CRITICAL TRAINING PHASE. If you are using this race as a tune-up for your primary ultras later in the summer, some VO2 max work leading up to the race is a good idea. Although you will not tap into that

intensity much, it will set the stage for those more important upcoming races. If this is your goal event, or your first ultra, you should be doing SteadyStateRun work in the four to eight weeks leading up to the race.

CREW TIPS. Thanks to the usually mild weather, crewing this race is easy. However, getting to the remote Warm Springs Creek aid station involves a lot more driving on winding roads than many expect. Crews will need to be self-contained and have a full tank of gas for the day because fuel stations and grocery stores are too far away to dash off for a resupply.

Pacers are not allowed on the course.

Leadville Trail 100

Held in late August in the Colorado Rockies, the Leadville Trail 100 is one of the oldest and most iconic 100-mile races in North America. Known for its high altitude (the lowest point of the race is 9,200 feet) and relatively large race field, the race attracts a number of first-time 100-mile runners to the starting line due to the fact that there is no qualifying requirement to enter.

Course record: 15:42:59 (Matt Carpenter, 2005); 18:06:24 (Ann Trason, 1994)

Median time: 28:20

Cutoff time: 30 hours

Climbing: Total elevation gain/loss: 18,168 feet

Significant climbs:

Colorado Trail and Hagerman Pass Road: 5.65 miles, 1,338 feet

Hope Pass: 4.48 miles, 3,204 feet

Sheep's Gulch: 2.4 miles, 2,303 feet

Leaving Twin Lakes to the Colorado Trail: 2.7 miles, 1,222 feet

Powerline: 3.88 miles, 1,483 feet

Significant descents:

Powerline: 3.88 miles, 1,483 feet

Colorado Trail into Twin Lakes: 2.7 miles, 1,222 feet

Sheep's Gulch: 2.4 miles, 2,303 feet

Hope Pass: 4.48 miles, 3,204 feet

Hagerman Pass Road and the Colorado Trail: 5.65 miles, 1,338 feet

DID YOU KNOW? Leadville local Ken Chlouber started the race in 1983 as a way to generate tourism revenue for the town, which at the time had the highest unemployment rate in the nation.

WEATHER. Typically cool, crisp mornings in the low 40s give way to mild afternoon temps in the mid-70s. Fast-moving thunderstorms roll through this area of the Rocky Mountains in the late afternoon, so if you are up high, especially above the timberline, make sure you have gear for nasty weather.

UNIQUE WEATHER-RELATED CHALLENGE. Late August can offer a dry weather window in the Colorado Rocky Mountain monsoon season, but every few years the race falls on the edge of the rainy season. If this is the case, expect heavy rain throughout the day and snow and sleet at higher elevations and during the night. In addition, between 10 p.m. and 4 a.m., it can be quite cold. Combine that with late-race fatigue and difficulty generating body heat, and you have a recipe for hypothermia. Miles 87 to 95 around Turquoise Lake exaggerate this particular constellation of calamities, so bring extra clothes for that section.

EQUIPMENT. Leadville is unique in the world of ultras in that pacers, who can join after mile 50, can mule (or carry the equipment) for their runners. The race directors instituted this rule as homage to the area's mining heritage and the miners' essential companion, the noble burro. Take advantage of this rule and have your pacer carry extra clothes, water, food, headlamp batteries, and trekking poles should you need them.

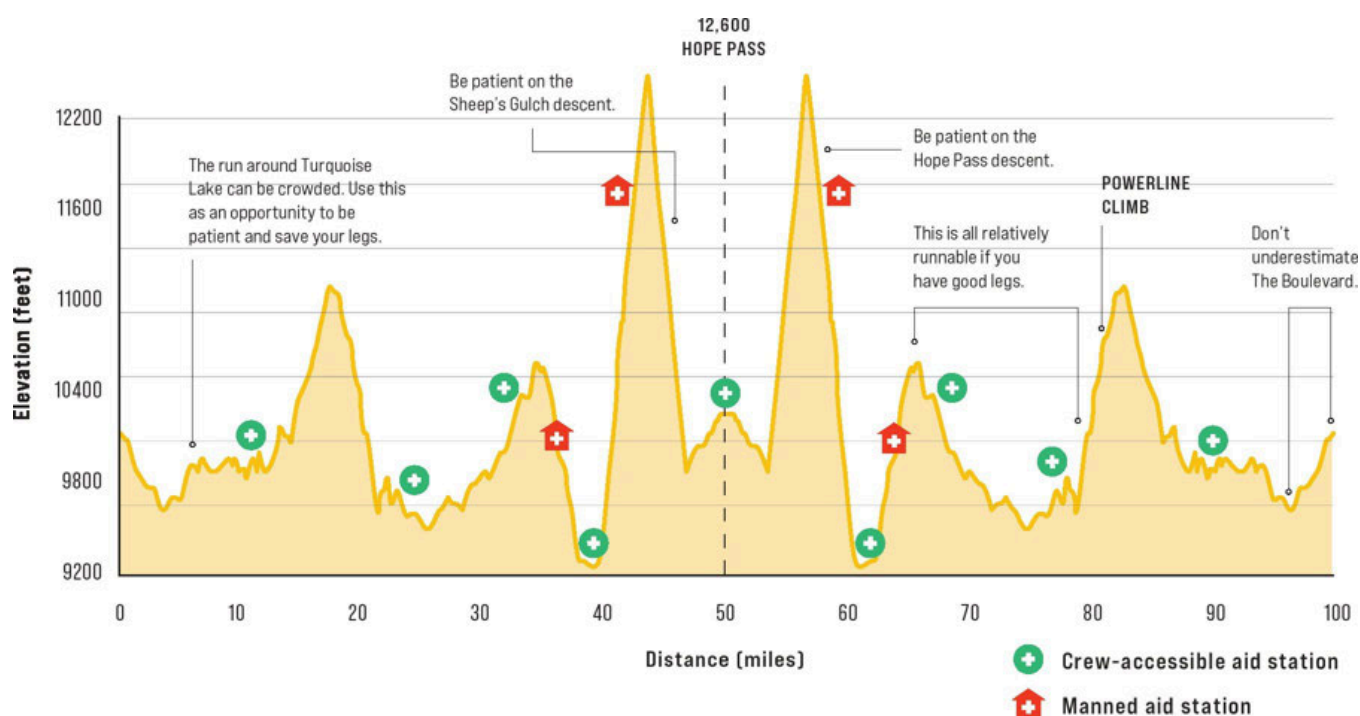
INDISPENSABLE GEAR. A pacer/mule to lighten your load.

CRUX OF THE RACE. Although the Hope Pass (12,600 feet) double crossing in the middle of the race always receives the most attention, the crux of the course is the climb up Powerline over Sugarloaf Pass (11,071 feet) and then the descent into the May Queen aid station at mile 86.5. This section comes late in the race, and the Powerline climb, although not long, is steep and slow-going. The section of the Colorado Trail leading into the May Queen aid station is one of the more technical parts of the course. Add in the darkness of night and tired legs, and it can be a rough go.

It can also be the weirdest: one of my most notable ultra hallucinations occurred on the Hagerman Road descent, where all the rocks came to life and were dancing and twirling across the gravel road!

CRITICAL MENTAL CHALLENGE. With as many as 600 to 800 participants and an out-and-back course, the race can become quite crowded. Many find this a source of inspiration, but it can also backfire as runners struggle to relax and run their own race among the crowd. Worse, some runners struggling to beat the cutoffs tend to cluster into groups of shared misery around miles 60 to 85, producing a negative feedback loop that slows everyone down to the pace of the slowest runner in the group. The result is that some runners who could have easily beaten the cutoffs do not. The key is to realize when you're in such a group and accelerate out of it immediately.

TRAINING TIPS. Although the course isn't particularly challenging from a technical perspective, many runners fear the altitude with good reason: it exaggerates any weakness in a runner's pacing and nutrition game plan. Thus, it's critical to practice and follow the hydration and nutrition plan you've mapped out for the race in the weeks and months leading up to the event. Come race day, slow and steady will pay off.



THE PRO KNOWS.

"There's a three-mile, gradual uphill finish into town called 'The Boulevard.' Coming at mile 97, it's more difficult than you'd expect. Don't think you are home free when you get to the Boulevard; continue pushing until you can see the finish line on 6th Street."

—Dylan Bowman, second place, 2011

CREW TIPS. With the start/finish in Leadville and the course staying in one general area, logistics for the race are relatively easy. But with so many runners bringing so many crew members to the race, and with few access points to aid stations, tempers among crews can flare over parking spots and positioning at the aid stations. There's no need for it; whether you're right next to the aid station check-in or thirty yards down the trail, it isn't going to matter. Tell them to chill out.

Tor Des Géants

Tor des Géants (simply known as the Tor) is the Super Bowl of 200-mile ultramarathons. Literally translated to "Tour of the Giants," the tough, rugged, remote route attracts runners from around the world wishing to take a crack at one of the most difficult endurance races on the planet. The route lies in the Aosta Valley of northwestern Italy and employs the notoriously steep Italian Alps, which athletes must climb, traverse, and descend over at least a few days' time. The route utilizes two of the oldest trails in the region (Alta Via 1 and Alta Via 2); connects more than thirty different municipalities, cities, and villages; and climbs twenty-five different mountain passes.

Distance: 221 miles

Median time: 133 hours

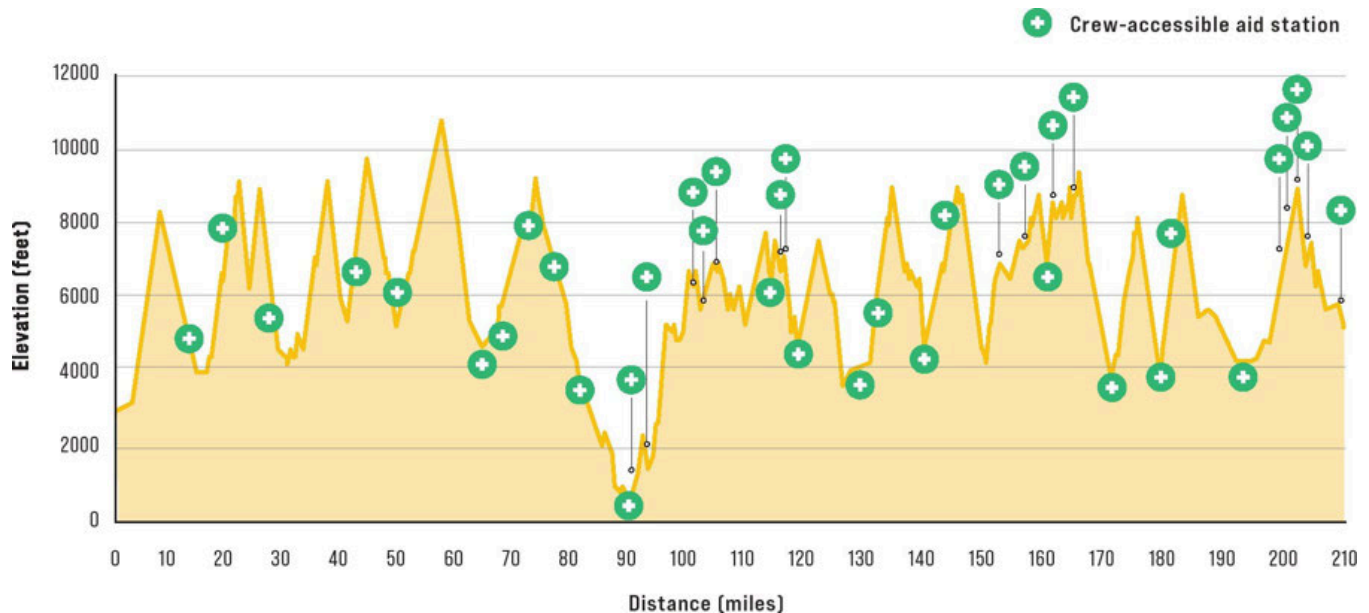
Cutoff time: 150 hours

Climbing: 82,000 feet of climbing and descending

DID YOU KNOW? The Tor des Géants brings out the entire valley to support the event. In the week leading up to and the week during the race, the 1,260-square-mile region becomes completely immersed in all things Tor. The small towns and villages the Tor passes through are adorned with Tor banners and billboards, and every shop seems to have some trinket adorned with the Tor logo, including postcards, special edition beers, and stuffed animals. The central town of Aosta has festivals, movie screenings, and athlete panels which runners, crew, and locals alike attend.

HOW TO GET IN. The Tor des Géants has one of the more interesting lottery processes: they weight the field by proportionally allocating entries based on the percentage of applicants by country. In other words, if 10 percent of the applicants hail from the United States, 10 percent of the 750-person field that is selected will be from the US. This makes the race field a true representation of the worldwide interest. There are no other prerequisites for entry.

WEATHER. Weather around the Aosta Valley can change dramatically from one mountain pass to the next. Be prepared for anything, literally. You can get hail and snow up one climb and have temps in the 80s on the following descent. Fortunately, the race produces "zone forecasts" for each of the five major geographic regions of the course. These can be accessed via an app or at the major aid stations (which are known as "Life Bases").



AID STATIONS. Aid stations at the Tor des Géants are a combination of high alpine refugios (small mountain cabins or huts) approximately every 10K and major aid stations called “Life Bases” every 50K. The Life Bases are literally a lifeline, offering runners warm meals, a shower, medical support, places to sleep, and access to your all-important yellow duffle bag (see Equipment section next). If you have support crew coming over to the Tor, consider yourself lucky. While the Aosta Valley is a great getaway, crewing for the Tor is a big favor to ask and normally chews up over a week of precious time. If you want crew, get your brownie points in order so that you can cash them in!

EQUIPMENT. Mandatory equipment is handled a bit differently at the Tor. When you receive your race bib, you will also receive a giant yellow duffel bag that will soon become your lifeline and best friend (feel free to give it a pet name, as well). Depending on the conditions, race management can require you to physically carry any of the equipment from the mandatory list or they can leave you to your own devices. At a minimum you must always carry cold- and foul-weather gear, a hat, gloves, two headlamps with backup batteries, your phone, and a passport. All of the mandatory gear required by the race and not physically on you while out on the course is to be placed in your yellow bag and returned to the race officials before the start. Pay attention, as the list is extensive and includes things like crampons, a knife, and extra string for a broken shoelace, all of which you might actually need. Similar to how the airlines treat luggage, your giant yellow bag follows you around the course from Life Base to Life Base and magically appears from a volunteer’s hands upon your arrival.

THE PRO KNOWS.

“Tor will test you like no other race. It is relentless. The climbs are long, hard, and seem to never end; and once you get through one of the high mountain passes, it’s back down the other side for a quad-smashing descent. But it’s all worth it. The return back to the start/finish line in Courmayeur will be one of, if not the most, memorable finishes of your ultra career. Whether or not you can remember the rest of the race is another question altogether. I only remember about 50 percent, so I will keep going back.”

—Jason Koop, 2019 finisher

INDISPENSABLE GEAR. Poles. Just look at the elevation statistics. Poles are an indispensable tool at the Tor. Get used to using them both uphill and downhill. Bring a backup pair to stow in your yellow gear bag should you break one (or both). A cylindrical cardboard tube normally used for architecture drawings and large maps is a good way to pack foldable poles to ensure they don't get crushed in transit.

CRITICAL LOGISTICAL CHALLENGE. Sleep. The winners will finish the Tor in a little over seventy hours. With enough caffeine, they might be able to make it through the race with just a trail nap or two. For the rest of the field, when and where you sleep is critical. Aid stations will limit you to two hours, and most of the time they are very strict on this. Take advantage of whenever you can get a full two hours of sleep regardless of whether it is in a high alpine refugio or Life Base. Refer back to Chapter 16 for sleep strategies.

TRAINING TIPS. Hike, hike, and hike some more. There is hardly any uphill running at the Tor. The race selection lottery for the Tor happens in February, meaning you will have about seven months to train from the time you know you are in the race to when the gun goes off. Use all of this time. You can spend February, March, and April in a normal running routine full of intervals and the like, but once May rolls around, your focus should be on low-intensity hiking.

CREW TIPS. Many families like to go to the Tor as a vacation of sorts. It's a great destination, as the area is magnificent and filled with adventures and history. If you do have support crew, have them stay in the central town of Aosta rather than where the race starts and ends in Courmayeur. This will save them about an hour's worth of driving per Life Base they go to, which will start to stack up as the days move along. Getting around from town to town is not all that difficult, but be mindful of the speed traps and paid parking lots. Italy is renowned for their traffic enforcement! You won't know if you've been caught speeding in a rental car until four months later when you get a bill in the mail. Trust me on this.

UTMB/CCC/TDS

The Ultra-Trail du Mont-Blanc (UTMB), Courmayeur-Champex-Chamonix (CCC), and Sur les Traces des Ducs de Savoie (TDS) are the three most popular races occurring over the UTMB weekend. While they do not all occur on exactly the same trails, the trails are similar enough that preparation for any of these races will be similar.

Median time:

UTMB: 40:46

CCC: 22:18

TDS: 39:27

Cutoff time:

UTMB: 46:30

CCC: 26:30

TDS: 44:00

Climbing:

UTMB: Total elevation gain/loss: 32,808 feet

CCC: Total elevation gain: 20,132 feet

Total elevation loss: 20,769 feet

TDS: Total elevation gain: 29,855 feet

Total elevation loss: 30,492 feet

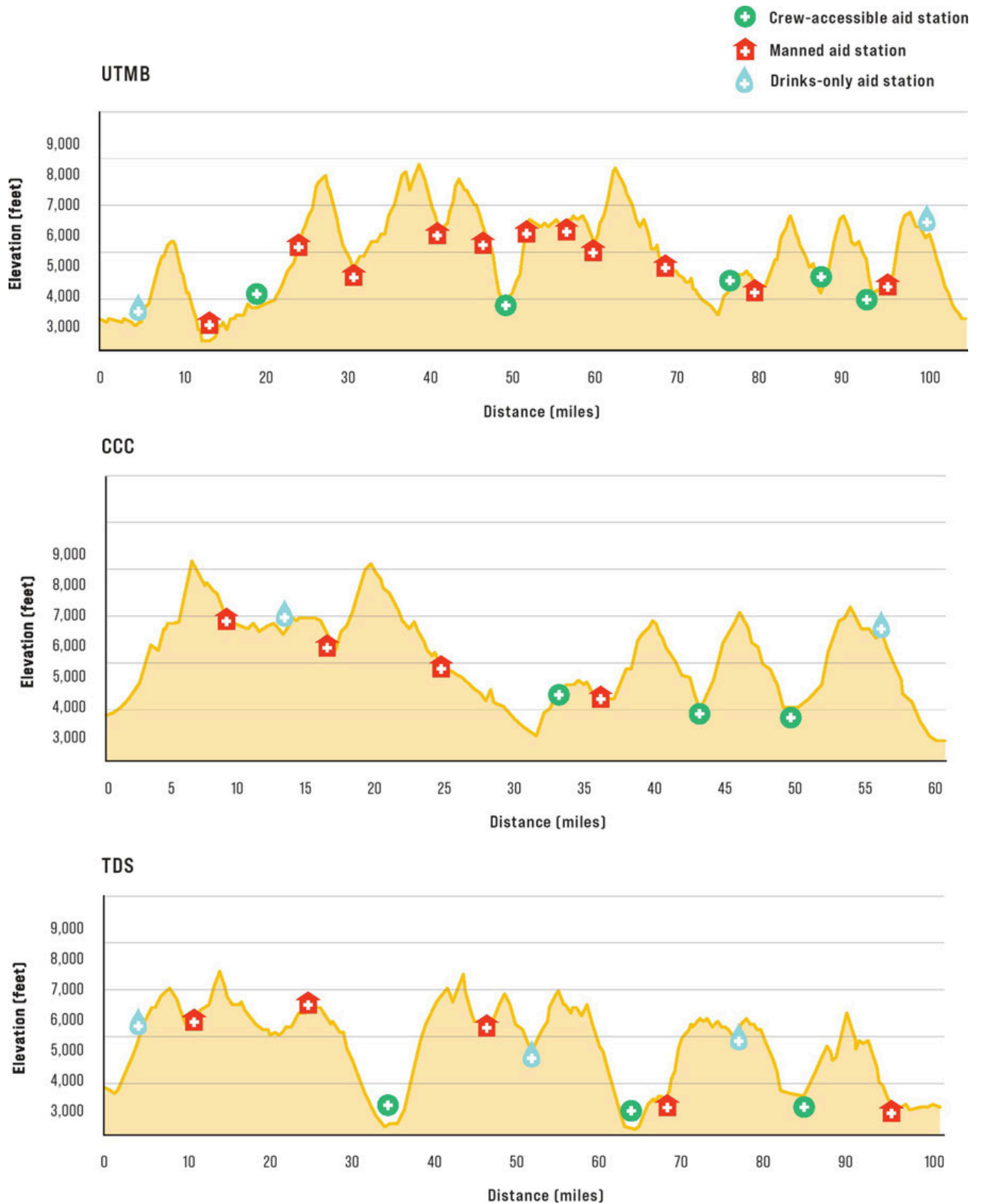
DID YOU KNOW? The UTMB races bring over 100,000 people and 10,000 runners to the Chamonix valley every year. The races are an enormous economic driver for the mountain town that sees all types of adventurers from BASE jumpers to backcountry skiers and, yes, ultrarunners. The UTMB races all revolve around the Mont Blanc massif, which spans the countries of France, Italy, and Switzerland.

HOW TO GET IN. The UTMB races are notoriously convoluted to enter. An elaborate system of qualifying points is necessary to apply for entry into each race. From there, the race organizers use a lottery system to fill each field. Every few years, different aspects of the points system change. Points are available in many countries, and more value is placed on longer races.

WEATHER. Weather around the Mont Blanc massif is notoriously moody. Particularly on the Italian side, clear and sunny skies can change to a hailstorm in an instant.

EQUIPMENT. The races all have mandatory gear that is periodically checked at aid stations. Don't be a weight weenie and try to skimp out on a more protective jacket or pants. Be prepared for bad weather if there is any hint of it in the weather forecast in the days leading up to the race.

INDISPENSABLE GEAR. Your pack. You must wear a pack and carry mandatory gear per the race rules. Train with it at least three days per week, even if you don't need all of the gear. Packs will ride differently depending on how they are loaded. Finding out that your mandatory headlamp jabs into your ribs when placed into a particular pocket is not something to save for race day.



THE PRO KNOWS.

“UTMB is a magical race. The energy of the race, spectators, and the mountain conjure up some intense emotions. So, try to get over to Chamonix a week in advance. That will give you time to adjust to the time difference and get

a couple days to relax before racing. Stay out of the heart of Chamonix. Staying just a mile or two up or down the valley is much quieter. The public transportation is great, so there's no need to get a car. Resist the urge to go explore the mountain once you arrive. The trails are stunning and hard to stay off, but curb your enthusiasm. Your legs will thank you on race day.

Play to your strengths. Are you good at descending? Flats? Uphills? UTMB is a race that really has it all. Know your strengths and weaknesses as a runner and use them to your advantage. Poles are a godsend, but make sure you know how to use them. Get your required gear ahead of time, and practice running with it.

Patience is key at UTMB. It's a long race, and hardly ever won over the first half of the race."

—*Stephanie Howe, three-time UTMB and onetime CCC finisher*

CRITICAL LOGISTICAL CHALLENGE. For UTMB, the evening start tends to throw off a lot of runners. Even for runners who get the opportunity to travel out to the race early and get adjusted to the time zone, sitting around all day twiddling your thumbs waiting for the start can be a gigantic waste of energy. Adding to the energy suck is the vibrancy of the town of Chamonix. With thousands of runners all converging on the small town at the same point in time, the city is a buzz of energy, which can drain the life out of a runner preparing to tackle the Mont Blanc. Staying slightly outside of the city is a good idea, as is ignoring all of your friends training too hard during their taper.

TRAINING TIPS. For many runners not accustomed to European trail running, the UTMB races will be a shock. The climbs are steep, and switchbacks are few and far between. Don't get lulled into a false sense of security by looking at the elevation gain/loss on paper; pay close attention to the grade and length of the individual climbs themselves. In North America, it is almost impossible to do too much climbing and descending to get prepared for this race. A training camp in the Rocky Mountains or Sierras would be a great addition to your program in the July or early August time frame. Additionally, if you do not run regularly with poles, get used to them. Using poles can be a big advantage at any of the UTMB races, and most runners would consider them indispensable.

CRITICAL TRAINING PHASE. July and early August should consist of a lot of low-intensity hiking on steep terrain (using your poles, of course). If you have done a good job building your intensity in earlier phases, you can get away with filling your last six weeks of training with EnduranceRuns (and hiking) and RecoveryRuns.

CREW TIPS. Many families like to go to UTMB with their runner as a vacation. Generally speaking, the race is very accessible to spectators yet difficult for crews, as there is a strict limit on when crew members can enter an aid station. Additionally, only one crew member is allowed to enter each station, so choose wisely. The race management will check for mandatory gear at aid stations, so crews should make sure their runner leaves with any mandatory equipment.

Vermont 100

One of the few 100-mile races on the East Coast and one of the oldest in North America, the Vermont 100 takes place every July on a large, looping course in eastern central Vermont. The well-marked, well-supported course follows country roads through bucolic, rolling farmland and New England forests. Unlike ultras out west, there are no sustained climbs or technical sections, but that doesn't mean it's easy. The oppressive humidity of a New England summer and the course can take their toll. Still, its easily accessed location, numerous aid stations and ample support, and lack of altitude make the Vermont 100 an attractive choice for many runners' first 100-miler.

Entrants to the 100-miler (there's a 100K race run on the same day) need to have completed a 50-mile ultra in under twelve hours or a 100K race in under fourteen hours, and to have completed a volunteer day at any ultra event that is 50K or longer.

Course record: 14:47:35 (Brian Rusiecki, 2014); 16:42:32 (Kami Semick, 2010)

Median time: 24:47

Cutoff time: 30 hours

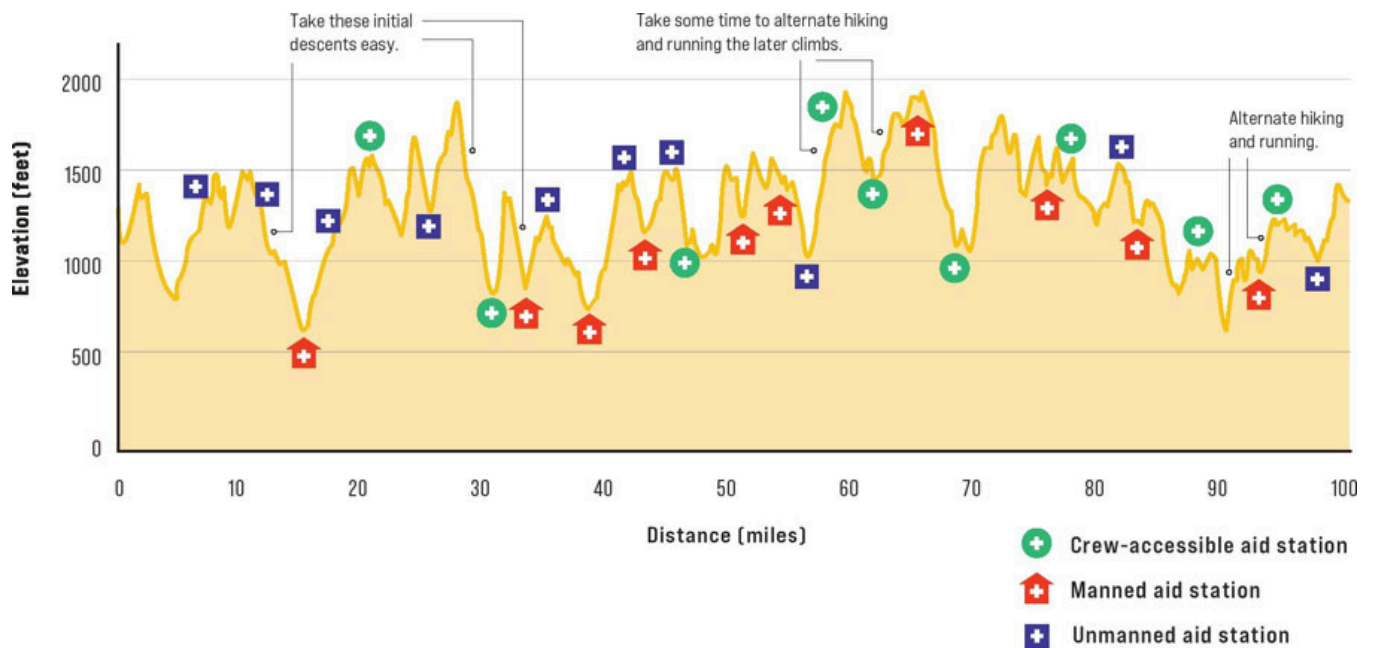
Climbing: Total elevation gain/loss: approximately 14,000 feet

DID YOU KNOW? The Vermont 100 is run congruently with a 100-mile endurance horse race. Runners share the course with the horses, with many of the equine finishing times equaling those of the top runners.

WEATHER. Summer temperatures in Vermont range from 80°F to 85°F in the day to 55°F to 65°F at night. As far as ultramarathons go, the conditions are quite pleasant. If you're lucky, you'll get a couple of late afternoon showers to cool things off quickly.

UNIQUE WEATHER-RELATED CHALLENGE. The high humidity will present a struggle for runners who are not acclimated to it. The body's ability to cool itself is hindered by the humidity, and runners will have to get used to running in sweat-drenched gear and deal with the potential for chafing that goes with it.

EQUIPMENT. Runners don't need to haul much with them. The relatively mild weather, easy-access route, and copious aid stations—the max distance between stations is five miles; the average distance between them is three and a half miles—mean that runners can get by with the bare necessities: a hat and a water bottle or two.



INDISPENSABLE GEAR. Cushioned road shoes, not trail shoes. Because the Vermont 100 travels along hard-packed gravel farm roads, jeep tracks, and short sections of pavement, it's a runnable course. Those dirt roads often surprise runners with how much pounding they dish out on their bodies. It's better to think of this event as a 100-mile road race rather than a trail run.

CRITICAL MENTAL CHALLENGE. On paper, the Vermont 100 doesn't look too hard—no giant climbs or technical sections—but the saw-edge course profile reveals a relentless series of short climbs and descents that never seem to end. It's difficult to find any sense of rhythm, the kind you might enjoy on the long climbs and descents of ultras out west. At first, running on country roads seems too easy, with sure footing, plenty of space to maneuver, and a well-marked course that's nearly impossible to get lost on. This causes many runners to start out too fast.

TRAINING TIPS. Heat training is vital, and if you can do it in a humid climate, so much the better. After that your priority should be preparing your quads for the relentless punchy downhill sections on a hard-packed surface. There's no need to worry about altitude; it's a nonfactor because the race tops out at under 2,000 feet in elevation.

THE PRO KNOWS.

"The crux of the race is the Camp Ten Bear Loop between mile 43 and mile 70. During this section the climbs and descents are longer, the tree cover is far less consistent, and the road surfaces are grittier. As a result, this innocent twenty-seven-mile section can take a lot out of a runner. If you're not careful through here you run the risk of a real death slog over the final thirty miles when you get more single-track, deeper tree cover, and the cool temperatures of the night."

—Andy Jones-Wilkins, three time Vermont 100 winner

CRITICAL TRAINING PHASE. Given that the race is generally rolling, expect to maintain a constant intensity, regardless of where you are in the pack. EnduranceRuns and SteadyStateRuns in the May–June time frame are advised, with your volume reaching its maximum in early June.

CREW TIPS. Because the course is a loop, you can camp out at the remote start/finish line the night before the race. This seems like it should be relatively easy, but keep in mind that the nearest town is thirty-five miles away. Arrive with everything you and your crew need for the race.

Follow the race bible to a T because it is very easy to get lost on the course. The race directors have it dialed to the 0.01th of a mile for a good reason: unmarked dirt roads head off in all directions, and your phone's GPS won't always work in this part of Vermont.

Wasatch Front 100

Held the weekend after Labor Day in September, this point-to-point race in Utah starts outside of Salt Lake City and traverses the Wasatch Mountains, heading southeast to finish in Soldier Hollow, Utah. Its unique feature is that the route, which runs between altitudes of 5,000 and 10,467 feet, stays high on the crest of the Wasatch, following various ridgelines, trails, and jeep roads until dropping down to the finish.

Applicants must complete eight hours of supervised trail maintenance work through a local Forest Service office and submit a verified work report to the race before being allowed to start.

DID YOU KNOW? The race ends at 5 p.m. on Saturday so that the vast majority of its volunteer force has plenty of time to get home in order to attend church on Sunday. The race also boasts one of the most professional and dedicated ham radio volunteer groups that tracks the runners throughout the race.

WEATHER. The Wasatch is known for extreme temperature swings, ranging from the mid-20s in the early morning to scorching mid-80s in the middle of the afternoon. Between mile 20 and mile 40, the dry heat kicks in.

UNIQUE WEATHER-RELATED CHALLENGE. Having the right gear to handle the 60-degree temperature range. You'll want ice sleeves and ice-filled bandanas to wear during the heat of the day, and double gloves, knit hat, and a down jacket for the nights in the mountains, where the technical trails don't allow you to move fast enough to generate sufficient heat to keep you warm. Some years have recorded snow on the mountain passes. Other years have seen temperatures top 100°F.

EQUIPMENT. Make sure you have enough gear to handle any weather conditions, from snow to triple-digit temperatures. Carry more water than you think you'll need, especially if you're at the front of the pack, because you'll be running through the hottest sections of the course at the hottest time of the day.

INDISPENSABLE GEAR. A packable down jacket that stuffs down to the size of a water bottle.

Course record: 18:30:55 (Geoff Roes, 2009); 22:21:47 (Bethany Lewis, 2014)

Median time: 31:32

Cutoff time: 36 hours

Climbing

Total elevation gain: 26,882 feet

Total elevation loss: 26,131 feet

Significant climbs:

Lamb's Canyon: 4 miles and 2,035 feet

Millcreek to Red Lovers Ridge: 6.5 miles and 2,000 feet

Catherine Pass: 3.6 miles and 1,761 feet

Significant descents:

Jeep road leading to the Francis Peak aid station: 4 miles and 1,715 feet

Bald Mountain to the Alexander Ridge aid station: 5.49 miles and 2,253 feet

Scott's Pass to Brighton Lodge: 3.14 miles and 1,135 feet

Catherine Pass to Ant Knolls aid station: 1.9 miles and 1,423 feet

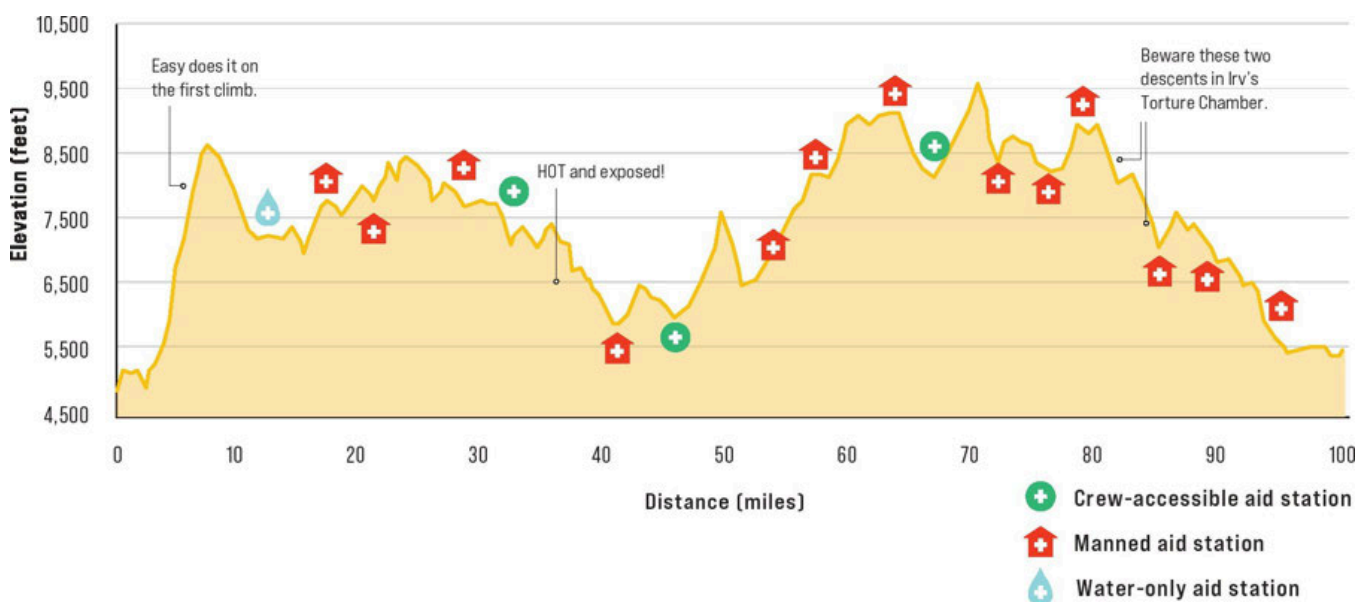
"The glide" and "the plunge": short, steep plunges starting at mile 79, known as "Irv's Torture Chamber"

CRUXES OF THE RACE. The descents off of Catherine Pass and "Irv's Torture Chamber" are rooted in ultrarunning lore for good reason. They occur late in the race, are tough, and will shred even the most seasoned runner's quads with steep technical descending.

CRITICAL MENTAL CHALLENGE. Getting out of the Brighton Lodge aid station at mile 67.3. It's just before the big climb up to the highest point of the race, Point Supreme, at 10,467 feet. Many runners arrive in the middle of the night, cold and wasted. The check-in is located right inside the A-frame, but all the

drop bags are fifty feet farther inside, where it's warm and cozy. Beyond that is a room referred to as "the morgue," so called because racers who enter it to rest often never leave. If you have a drop bag inside, grab it and then head immediately back outside, or have your crew wait for you outside to help you. Whatever you do, don't spend any more time inside than you have to.

TRAINING TIPS. Depending on your fitness, you can run approximately 70 percent of the course, much of which follows smooth roads and singletrack. That said, don't be lulled into thinking that it's easy. The relentless climbing and descending will take their toll, and you'll want to train your body accordingly. If running in heat is a known weakness, train for it by running in the hottest part of the day for some of your runs and perhaps using a sauna for additional heat acclimation. And although the altitude is not as extreme as in races in Colorado, it tops out in the last third of the race and hits most runners in the early hours of the morning.



THE PRO KNOWS.

"The climb out of the Brighton aid station is one of lore. But in reality, it's really not that hard. The combination of the climb being late in the race, normally at night, and after an easy road descent will make you feel like it is harder than it actually is. When you get to the Brighton aid station, get your food and gear and get out!"

—Jason Koop, *eleventh place, 2012*

CRITICAL TRAINING PHASE. July and August you should be doing big EnduranceRun miles, lots of climbing and SteadyStateRun intensity.

CREW TIPS. Pacers can join runners at mile 45. Pacers running through the night, and especially those starting out of the Brighton Lodge aid station, should be dressed for hiking—not running—in subfreezing temperatures.

Western States 100

The granddaddy of North American ultramarathons, the Western States 100-Mile Endurance Run was first run officially in 1977. The point-to-point trail race starts in Olympic Valley, California, crosses the Sierra Nevada, and descends to the finish in Auburn, California. Its heritage and global prestige attract the world's best ultrarunners to the race, making this arguably the most competitive ultra race in North America, if not the world. A win here can make a career.

Competitors are chosen by lottery from a pool of those who've completed either a 100-mile or another qualifying ultra race within a year's time (November to November). The sub-100-mile qualifiers come with time cutoffs determined by Western States race officials.

Course record: 14:09:28 (Jim Walmsley, 2019); 16:47:19 (Ellie Greenwood, 2012)

Median time: 26:03

Cutoff time: 30 hours

Climbing:

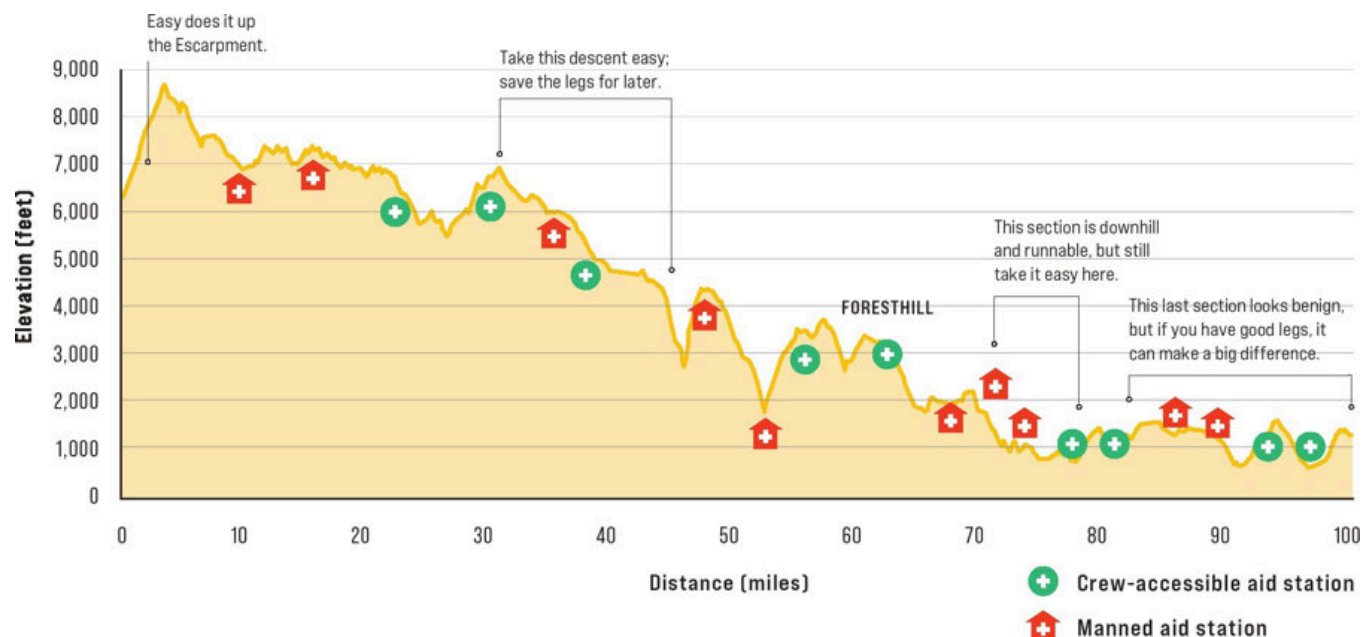
Total elevation gain: 18,090 feet

Total elevation loss: 22,970 feet

DID YOU KNOW? The Western States 100-Mile Endurance Run originally began as a 100-mile endurance horse race, which was called the Tevis Cup, after several local horsemen completed the route in 1955. In 1974, Gordy Ainsleigh, a veteran Tevis Cup rider, decided to run the race just to see if he could, finishing in 23:42. In 1977, the run separated from the trail ride and became the first official Western States Endurance Run.

WEATHER. The race is held on the last full weekend in June. The timing, close to the summer solstice, means a short night of chilly temperatures in the high 20s and long days in the hot sun, with temperatures well into the 90s. Temperatures over 100°F are not uncommon. Even at altitudes of 6,000 to 8,000 feet, the heat will make its presence felt. Rain is unlikely, but it can occur.

UNIQUE WEATHER-RELATED CHALLENGE. Fast runners will encounter the most exposed section of the course—with no shade and the possibility of temperatures pushing 110+°F—during the hottest part of the day, between 2 p.m. and 5 p.m. But it's not just fast runners who need to worry; heat is cited as an issue across the pack.



EQUIPMENT. Due to the heat, bandanas, headbands, a hat, or ice sleeves that can be filled with ice will go a long way toward making your day tolerable. But thanks to the high number of aid stations (twenty-one), you don't necessarily need to haul extra gear beyond fluids.

INDISPENSABLE GEAR. A bandana you can soak in ice water, fill with ice, and wrap around your head or neck to cool you down.

CRITICAL MENTAL CHALLENGE. The roughly seventeen-mile descent down the Cal Street section out of Foresthill School, which starts at mile 62, can hit runners during the hottest parts of the day. The relatively gradual decline drops more than 3,000 feet, and it's deceptively easy to go too fast here and blow out your quads. You need to be conscious of when it feels easy during this stretch and hold your speed in check. That way, you'll have enough in your legs to get through the punchy climbs in the last twenty miles.

TRAINING TIPS. Acclimate to running in heat by doing some of your training runs during the hottest parts of the day, making sure to experiment with your fluid and nutrition intake to find out what works. If you can, take advantage of access to a sauna in the last several days leading up to the race. This will help you get used to the heat. In fact, this is so important that one of my colleagues has taken a thermal imaging camera out to the race to analyze runners' temperatures. Her results? The cooler runners fared better than the hotter ones.

THE PRO KNOWS.

“Every ultrarunner has this race on their bucket list. As a result, you end up with a very competitive international field. If you’re not careful, you can get caught up in the competition, reacting to other racers’ moves and eventually blowing yourself apart instead of sticking to your own race strategy.”

—*Dylan Bowman, three-time finisher, third place in 2014*

CRITICAL TRAINING PHASE. April and early May with TempoRun work. Western States is one of the few races where I insist that runners find a dry sauna to help them acclimate to the heat. Therefore, it is critical that their cardiovascular fitness is built to a maximum before the sauna protocol starts.

CREW TIPS. Although there are a number of aid stations open to crews, some require shuttles. And because it’s a point-to-point race through remote stretches of the Sierra Nevada, crew members spend a lot of time driving, often on twisting, narrow mountain roads.

Pacers can be used starting at the Foresthill aid station at mile 62.

